IoT in Smart Agriculture

Sensor-Based Irrigation as a Practical Application

INTERNSHIP RESEARCH REPORT

Focus: How connected sensors can support irrigation decisions, and what is needed for the approach to work on real farms.

Abstract

This report examines how the Internet of Things (IoT) can support smart agriculture, focusing on sensor-based irrigation. In this approach, field devices measure soil and weather conditions, send readings to a local controller or online service, and help a farmer decide when and where to irrigate. The technology can improve the timing of water application and make field conditions easier to observe. Its value, however, depends on reliable measurements, suitable connectivity, maintenance and farmer involvement. A review of technical literature and documented implementation evidence shows that IoT is best understood as a decision-support system, rather than a device that automatically solves water scarcity.

1. Introduction

Agriculture is increasingly expected to produce dependable harvests while managing water, energy and soil carefully. Fixed schedules cannot account for every change in rainfall, soil, crop growth or temperature, so some areas may receive too much water while others remain dry.

The Internet of Things (IoT) connects devices that collect and exchange data. Farm devices include soil-moisture probes, weather sensors, flow meters and valves. This report reviews smart irrigation, its benefits, limits and conditions for responsible implementation. It draws on peer-reviewed reviews, an FAO study of 22 digital-agriculture cases, and a World Bank pilot description.

2. How an IoT irrigation system works

A practical system has four linked parts. First, sensors collect measurements. Soil-moisture probes estimate how much water is available near the roots; a weather station may record temperature, humidity, rainfall and wind; and a flow meter can show how much water passes through a pipe. Placement matters: one probe cannot represent an uneven field, and soil texture affects readings.

Second, a microcontroller or gateway gathers the readings. It can apply a simple rule, such as alerting the farmer when moisture falls below a crop-specific threshold. Advanced tools combine soil data with crop stage, forecasts and estimated water demand. Third, the system transmits readings over Wi-Fi, cellular networks or low-power radio technologies such as LoRaWAN. A dashboard or mobile application can display trends and send alerts. In locations with weak internet, a local controller may store data and continue operating offline.

Finally, a farmer or controller acts on the information. Irrigation can be started manually or by a controller operating a valve or pump. A safer design includes limits, such as maximum run time, leak detection and a manual override. Automation should be calibrated to local crops and soil rather than using one universal moisture threshold. Together, these elements form a feedback loop: measure, interpret, irrigate and measure again.

3. Applications and potential benefits

IoT irrigation schedules watering from field conditions rather than a fixed calendar. Avoiding unnecessary watering can lower water use, pumping costs and field-checking time. Alerts can flag pump or line problems early.

Evidence supports potential, while cautioning against exaggerated claims. García et al. (2020) describe common sensors, controllers, communications and implementation challenges in IoT irrigation. Abdelmoneim and colleagues' 2025 systematic review reports growing research interest and a move toward low-power communication and integrated decision support. These reviews describe available approaches; they do not mean every farm will achieve the same savings. FAO's analysis of 22 cases similarly concludes that digital and automation tools can improve efficiency and sustainability, while emphasizing that economic, environmental and social effects need stronger evidence.

4. Real-world relevance and implementation challenges

A relevant example is the World Bank's Knowledge for Change Program pilot on IoT sensors for irrigation efficiency. Its FY2024 annual report describes work with 137 smallholder farmers in India and Vietnam. The pilot explored using IoT to help implement Alternate Wetting and Drying (AWD), a rice irrigation approach that allows fields to dry to a suitable level before being flooded again. This links the technology to a specific water-management practice and to farmers who act on its signals. As a pilot, it is not proof of universal impact or commercial success.

Cost is a major barrier. Sensors, gateways, subscriptions, solar power, installation and repair can make a system unaffordable, particularly for small farms. Connectivity and electricity may be intermittent. Sensors drift, corrode or fail; poor placement can produce misleading readings. Different vendors may use incompatible formats, and cloud services can create dependence on a supplier. Data ownership and privacy also matter if farm information is stored or shared without clear terms. A system that requires specialist support unavailable locally is unlikely to remain useful after a demonstration ends.

FAO identifies cost, skills, infrastructure and an enabling policy environment among barriers to adoption. It highlights rural internet and electricity investment, digital literacy, adaptation to different farm types and collection of better impact evidence. These observations suggest that successful deployment is partly organizational: training, repair access, clear alerts and farmer control belong in the design alongside sensors and software.

5. Recommendations and conclusion

A responsible implementation should begin with a clearly defined problem, such as deciding when to irrigate one crop block. The user should then select robust, locally supportable sensors and test them against field observations. Data should be checked for missing or implausible readings, and irrigation

rules should be reviewed by an agronomist or experienced farm adviser. Farmers need training and a manual override. Where possible, the system should continue basic operation without internet, use secure access controls, and allow data export in a usable format.

Evaluation should compare the IoT-assisted method with the farm's usual practice across a full crop cycle. Useful measures include water applied, energy use, labor time, crop yield and quality, equipment downtime, repair cost and farmer satisfaction. Reporting these measures together helps reveal trade-offs; saving water while reducing yield may not be a good outcome.

IoT-based smart irrigation offers a practical way to make farm conditions more visible and irrigation decisions more responsive. Sensors, communication links, decision tools and valves can work as one system, but the result depends on fit with local conditions and on people being able to maintain and use it.

Current evidence is promising but varies by setting, and pilots should not be presented as universal proof. For agriculture interns, the main lesson is that effective IoT work combines technical design with agronomy, field validation, affordability and farmer participation.

References

Sources consulted for the report. URLs and DOI links are provided for verification.

1. Abdelmoneim, A. A., Kimaita, H. N., Al Kalaany, C. M., Derardja, B., Dragonetti, G., & Khadra, R. (2025). IoT Sensing for Advanced Irrigation Management: A Systematic Review of Trends, Challenges, and Future Prospects. Sensors, 25(7), 2291. https://doi.org/10.3390/s25072291

2. Ceccarelli, T., Chauhan, A., Rambaldi, G., Kumar, I., Cappello, C., Janssen, S., & McCampbell, M. (2022). Leveragingautomation and digitalization for precision agriculture: Evidence from the case studies. FAO Agricultural Development Economics Technical Study No. 24. https://doi.org/10.4060/cc2912en

3. García, L., Parra, L., Jimenez, J. M., Lloret, J., & Lorenz, P. (2020). IoT-Based Smart Irrigation Systems: An Overview on theRecent Trends on Sensors and IoT Systems for Irrigation in Precision Agriculture. Sensors, 20(4), 1042. https://doi.org/10.3390/s20041042

4. World Bank. (2024). Knowledge for Change Program: Annual Report FY24. See the project description, “IoT in Agriculture –Sensors Application for Irrigation Efficiency.” https://thedocs.worldbank.org/en/doc/acf5d8bcce6f4d80c6a47166404b2881-0060052024/original/KWPF-Annual-Report-FY24-Final-Digital.pdf

Prepared as a research-based internship assignment report. Add your own name, institution and internship details if your submission format requires them.